

Week 1 (Proposal)

I contributed to dataset understanding and structure. I described that the dataset contains 303 rows and 14 columns, including medical features such as age, cholesterol, and chest pain type.

Week 2 (EDA & Preprocessing)

I implemented the main exploratory data analysis and initial inspection.

Dataset loading:

```
from ucimlrepo import fetch_ucirepo

heart_disease = fetch_ucirepo(id=45)

df = heart_disease.data.features.copy()

df['target'] = heart_disease.data.targets
```

Basic inspection:

```
print(df.head())

print(df.info())

print(df.describe())

print(df.isnull().sum())
```

From this, I identified:

Dataset shape: (303, 14) Missing values: ca (4) and thal (2)

Class distribution:

```
import seaborn as sns

sns.countplot(x='target', data=df)
```

Insight:

Most patients do not have heart disease Among diseased, most cases are mild

Histograms:

```
df.hist(figsize=(12, 10))
```

Insight:

Most patients are between 40–60 years old

Boxplot:

```
sns.boxplot(x='target', y='thalach', data=df)
```

Insight:

Higher disease → lower heart rate (thalach decreases)

Correlation heatmap:

```
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

Insight:

Strongest predictors: cp and thalach

Scatter plot:

```
sns.scatterplot(x='age', y='chol', hue='target', data=df)
```

Insight:

Older age + high cholesterol → higher disease risk Week 3 (Modeling)

I supported modeling by providing insights from EDA to guide feature selection and model interpretation.

```
import matplotlib.pyplot as plt
```

```
from ucimlrepo import fetch_ucirepo
```

```
heart_disease = fetch_ucirepo(id=45)
```

```
df = heart_disease.data.features.copy()
```

```
df['target'] = heart_disease.data.targets
```

Week 3 (Modeling)

I implemented the K-Nearest Neighbors (KNN) model to capture complex, non-linear patterns that the baseline model might have missed. After training the model and evaluating its performance, I utilized Pandas to integrate its results into a comparison table. I also conducted cross-validation to verify that the model's high performance was consistent across different subsets of the data.

```
from sklearn.neighbors import KNeighborsClassifier
```

```
knn = KNeighborsClassifier(n_neighbors=5)
```

```
knn.fit(X_train_scaled, y_train)
```

- KNN achieved the highest overall accuracy and precision.

FINAL BLOCK

This form is approved and signed by all team members:

Alikhan Sagat

Zhaslan Karakat

Kenzhegaliyev Nursultan

Azizbek Ainaz